LAB SHEET 6

Introduction to Mininet-WiFi: Stations & Access Points

Week	6
Duration	2 hours
Deliverable	wifi_topo.py, screenshot report (PDF), pushed to your netprog-2026 GitHub repository

Learning Objectives

By the end of this lab, you should be able to:

1. Explain the architectural differences between Mininet and Mininet-WiFi (stations vs. hosts, access points vs. switches, and the role of wmediumd in emulating the wireless medium).
2. Write a Mininet-WiFi topology using the Python API: addStation(), addAccessPoint(), and configureWifiNodes().
3. Configure basic signal-related parameters on stations and access points (position and range).
4. Verify wireless association and connectivity between stations via an access point.
5. Interpret association/signal information using Mininet-WiFi-specific CLI and shell commands (iw).

Prerequisites

- Lab 1 completed: Mininet-WiFi installed and verified with the simplewifi.py example.
- Lab 5 completed (comfort with the Topo-free, script-based Mininet Python API pattern: build a net object, call .start(), use CLI(net), call .stop()).

Software & Materials

- Mininet-WiFi (installed in Lab 1), Python 3.8+
- A text editor / IDE
- iperf/iperf3

Lab Tasks

Complete the following tasks and record evidence (terminal output / screenshots) for each in your report.

Task 1: Screenshot of Part B's minimal topology with net.pingAll() reporting 0% dropped.

```
osboxes@osboxes:~/netprog-2026/lab06_wifi_intro$ sudo python3 wifi_topo.py
*** Creating nodes
*** Configuring wifi nodes
*** ap1-wlan1: minimum tx power (1 dBm) yields 15.00m for requested 15.00m (delta +0
.00m)
*** Starting network
*** Starting controller(s)
*** Starting L2 nodes
ap1 ...
*** Ping: testing ping reachability
sta1 -> *** sta1 : ('ping -c1  10.0.0.2',)
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1032 ms

--- 10.0.0.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1032.319/1032.319/1032.319/0.000 ms
sta2
sta2 -> *** sta2 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.70 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.704/1.704/1.704/0.000 ms
sta1
*** Results: 0% dropped (2/2 received)
*** Starting CLI:
mininet-wifi>
```

Task 2: Screenshot of iw dev sta1-wlan0 link (Part C) showing sta1 associated with ap1, including SSID and signal information.

```
--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.507/1.507/1.507/0.000 ms
sta1
*** Results: 0% dropped (2/2 received)
mininet-wifi> net
sta1 sta1-wlan0:wifi
sta2 sta2-wlan0:wifi
ap1 ap1-wlan1:wifi ap1-wlan1:wifi
mininet-wifi> sta1 iw dev sta1-wlan0 link
Connected to 02:00:00:00:02:00 (on sta1-wlan0)
        SSID: netprog-ap
        freq: 2412.0
        RX: 29609 bytes (625 packets)
        TX: 1752 bytes (18 packets)
        signal: -49 dBm
        rx bitrate: 54.0 MBit/s
        tx bitrate: 54.0 MBit/s
        bss flags: short-slot-time
        dtim period: 2
        beacon int: 100
mininet-wifi> sta2 iw dev sta2-wlan0 link
Connected to 02:00:00:00:02:00 (on sta2-wlan0)
        SSID: netprog-ap
        freq: 2412.0
        RX: 37881 bytes (811 packets)
        TX: 1752 bytes (18 packets)
        signal: -49 dBm
        rx bitrate: 54.0 MBit/s
        tx bitrate: 54.0 MBit/s
        bss flags: short-slot-time
        dtim period: 2
        beacon int: 100
mininet-wifi>
```

Task 3: Screenshot of the iperf sta1 sta2 result from Part D.

```
osboxes@osboxes:~/netprog-2026/lab06_wifi_intro$ sudo python3 wifi_topo.py
*** Creating nodes
*** Configuring wifi nodes
*** ap1-wlan1: minimum tx power (1 dBm) yields 15.00m for requested 15.00m (delta +0
.00m)
*** Starting network
*** Starting controller(s)
*** Starting L2 nodes
ap1 ...
*** Ping: testing ping reachability
sta1 -> *** sta1 : ('ping -c1  10.0.0.2',)
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1018 ms

--- 10.0.0.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1018.148/1018.148/1018.148/0.000 ms
sta2
sta2 -> *** sta2 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=3.45 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 3.451/3.451/3.451/0.000 ms
sta1
*** Results: 0% dropped (2/2 received)
*** Starting CLI:
mininet-wifi> iperf sta1 sta2
*** Iperf: testing TCP bandwidth between sta1 and sta2
*** Results: ['10.0 Mbits/sec', '10.0 Mbits/sec']
mininet-wifi>
```

Task 4: Screenshot of Part E's pingall and iw dev sta3-wlan0 link BEFORE moving sta3 into range, showing sta3 failing to associate/ping.

```
x
sta2 -> *** sta2 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.23 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.227/2.227/2.227/0.000 ms
sta1 *** sta2 : ('ping -c1  10.0.0.3',)
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
From 10.0.0.2 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.3 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms

x
sta3 -> *** sta3 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
From 10.0.0.3 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms

x *** sta3 : ('ping -c1  10.0.0.2',)
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
From 10.0.0.3 icmp_seq=1 Destination Host Unreachable

--- 10.0.0.2 ping statistics ---
1 packets transmitted, 0 received, +1 errors, 100% packet loss, time 0ms

x
*** Results: 66% dropped (2/6 received)
mininet-wifi> sta3 iw dev sta3-wlan0 link
Not connected.
mininet-wifi>
```

With sta3 is located 40 meters away from the AP, which exceeds the range of 15.

Task 5: Screenshot of the same two checks AFTER moving sta3 within range, showing successful association

```
rtt min/avg/max/mdev = 2.275/2.275/2.275/0.000 ms
sta3
sta2 -> *** sta2 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.79 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.787/1.787/1.787/0.000 ms
sta1 *** sta2 : ('ping -c1  10.0.0.3',)
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=3.61 ms

--- 10.0.0.3 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 3.607/3.607/3.607/0.000 ms
sta3
sta3 -> *** sta3 : ('ping -c1  10.0.0.1',)
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.41 ms

--- 10.0.0.1 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.407/1.407/1.407/0.000 ms
sta1 *** sta3 : ('ping -c1  10.0.0.2',)
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.58 ms

--- 10.0.0.2 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 1.582/1.582/1.582/0.000 ms
sta2
*** Results: 0% dropped (6/6 received)
```

```
mininet-wifi> sta3 iw dev sta3-wlan0 link
Connected to 02:00:00:00:03:00 (on sta3-wlan0)
        SSID: hetprog-ap
        freq: 2412.0
        RX: 373621 bytes (8244 packets)
        TX: 2888 bytes (39 packets)
        signal: -49 dBm
        rx bitrate: 54.0 MBit/s
        tx bitrate: 1.0 MBit/s
        bss flags: short-slot-time
        dtim period: 2
        beacon int: 100
mininet-wifi>
```

In the red frames, they show pingall with result as 0% dropped and connect on sta3. And currently 10 meters away from the AP.

Part F: Mininet vs. Mininet-WiFi, and Capturing Real Frames

```
mininet-wifi> sta2 tcpdump -i sta2-wlan0 -c 6 &
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on sta2-wlan0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:44:47.091713 IP6 fe80::ff:fe00:100 > ip6-allrouters: ICMP6, router solicitation, length 16
12:44:47.603021 IP6 fe80::ff:fe00:0 > ip6-allrouters: ICMP6, router solicitation, length 16
12:44:49.652144 IP6 fe80::ff:fe00:200 > ip6-allrouters: ICMP6, router solicitation, length 16
12:44:59.033890 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 9770, seq 1, length 64
12:44:59.034573 IP 10.0.0.2 > 10.0.0.1: ICMP echo reply, id 9770, seq 1, length 64
12:45:00.035135 IP 10.0.0.1 > 10.0.0.2: ICMP echo request, id 9770, seq 2, length 64
6 packets captured
7 packets received by filter
0 packets dropped by kernel
mininet-wifi> sta1 ping -c 5 sta2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.06 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.22 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.38 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=1.51 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=1.47 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.215/1.527/2.058/0.284 ms
mininet-wifi>
```

I captured traffic on sta2-wlan0 while sta1 pinged sta2. The capture shows ICMP echo requests from 10.0.0.1 to 10.0.0.2 and echo replies in the opposite direction. All five pings succeeded with 0% packet loss, confirming connectivity between the stations through their simulated Wi-Fi association with the AP.

Task 6: In 2-3 sentences, describe one architectural difference between a Mininet switch and a Mininet-WiFi access point, and one difference between a host and a station, referencing something you observed today.

Answer In Mininet, a switch forwards traffic over simulated wired links, while a Mininet-WiFi access point connects stations through a wireless association. I could see this in the lab: sta1 used sta1-wlan0, and iw dev sta1-wlan0 link showed its connection to ap1, a regular Mininet host would use a wired interface such as eth0.

Verification Checklist

#	Checkpoint	From	✓
1	Minimal AP + 2-station topology (Part B) runs and pingall reports 0% dropped	Task 1	☑
2	iw dev sta1-wlan0 link (Part C) shows a successful association with the correct SSID	Task 2	☑
3	iperf sta1 sta2 (Part D) reports a nonzero bandwidth figure	Task 3	☑
4	Out-of-range sta3 (Part E, before moving) fails to associate and fails pingall	Task 4	☑
5	In-range sta3 (Part E, after moving) associates successfully and passes pingall	Task 5	☑
6	tcpdump capture (Part F) shows real ICMP frames on a station's wlan interface	After task 5, in part F topic	☑

Reflection Questions

Answer briefly (2-4 sentences each) in your report:

1. **ECompare a Mininet switch and a Mininet-WiFi access point: what does each actually emulate at the OS level, and why can't a plain Mininet switch provide wireless connectivity?**

   Answer A Mininet switch connects traffic between virtual Ethernet ports. A Mininet-WiFi access point connects stations through simulated Wi-Fi, but stations first connect to it through a simulated wireless association. A normal switch cannot let stations join a Wi-Fi network.

2. **Explain what net.configureWifiNodes() does and why it must be called after all addStation()/addAccessPoint() calls but before net.build().**

   Answer configureWifiNodes() sets up the wireless interfaces for the stations and access points. It must run after all the nodes are added so it can configure every interface, and before net.build() so the wireless network is ready when it starts.

3. **Why did the out-of-range station in Part E fail to associate, and what real-world wireless phenomenon does the range parameter approximate?**

   Answer When sta3 was at 90,50,0, it was 40 metres from ap1, beyond the AP's 15-metre range, so it could not associate or ping the other stations. Moving it to 60,50,0 put it within range and restored connectivity. The range setting approximates the coverage limit of a real Wi-Fi signal

End of Lab Sheet 6